

POLITECNICO DI MILANO • DETAILED COURSE SYLLABUS

Academic Prerequisites & Undergraduate Course Descriptions for MSc in Computer Science and Engineering

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Conferring Institution: St. Mary's University (Faculty of Informatics), Addis Ababa, Ethiopia • Degree: BSc in Computer Science (4-Year, Full-Time, 100% English MOI)

1. CORE COMPUTER SYSTEMS, ARCHITECTURE & OPERATING SYSTEMS

Operating Systems Architecture (CoSc 3011)

Credits: 4 ECTS Equiv: 7 ECTS

Category: Operating Systems & Low-Level Computing | **Lab:** 3 hrs/wk | **Lecture:** 3 hrs/wk

Comprehensive study of multi-programming operating system architectures, kernel design, process management, and concurrency.

- **Topics Covered:** Process scheduling algorithms (round-robin, multi-level feedback queues), thread lifecycles, POSIX threads, inter-process communication (IPC), pipes, shared memory.
- **Synchronization & Concurrency:** Race conditions, critical sections, semaphores, mutex locks, monitors, deadlock conditions, Banker's algorithm, prevention and recovery.
- **Memory & File Systems:** Virtual memory, paging, segmentation, page fault handling, page replacement (LRU, clock algorithm), VFS, inode management, disk scheduling.
- **Practical Lab:** C/Linux programming for kernel system calls, fork-exec routines, POSIX synchronization, and custom memory management routines.

Computer Organization and Architecture (CoSc 2032)

Credits: 4 ECTS Equiv: 6 ECTS

Category: Hardware-Software Interface & Architecture | **Lab:** 2 hrs/wk | **Lecture:** 3 hrs/wk

Fundamental concepts of digital logic, microprocessor architecture, instruction set architectures, and memory hierarchies.

- **Topics:** Von Neumann architecture, CPU datapath and control unit design, ALU operations, instruction pipelining, pipeline hazards (data, control, structural), branch prediction.
- **Memory Systems:** Cache mapping techniques (direct, set-associative, fully associative), cache coherence, write policies, virtual memory translation, bus protocols, DMA controllers.
- **Instruction Sets:** Assembly language programming (MIPS/x86), register allocation, stack frame manipulation, hardware interrupt servicing.

Distributed Systems (CoSc 4011)

Credits: 3 ECTS Equiv: 6 ECTS

Category: Distributed Systems & Scalable Architectures | **Lab:** 2 hrs/wk | **Lecture:** 3 hrs/wk

Architectures, theoretical models, and implementation principles for networked distributed and decentralized computing environments.

- **Foundations:** Characterization of distributed systems, client-server vs peer-to-peer topologies, remote procedure calls (RPC), message passing interfaces, network socket programming.
- **Time & State:** Physical and logical clocks (Lamport timestamps, vector clocks), causal ordering, global state snapshot algorithms.
- **Coordination & Fault Tolerance:** Distributed mutual exclusion, leader election (Bully, Ring), consensus protocols, replication models (primary-backup, active replication), Paxos/Raft overview, Byzantine fault models.

2. ALGORITHMS, DATA STRUCTURES & SOFTWARE ENGINEERING

Data Structures and Algorithms (CoSc 2011)

Credits: 4 ECTS Equiv: 7 ECTS

Category: Foundational Computer Science | **Lab:** 3 hrs/wk | **Lecture:** 3 hrs/wk

Rigorous analysis and implementation of abstract data types, recursive procedures, and algorithmic complexity.

- **Structures:** Linked lists, stacks, queues, binary search trees, AVL trees, Red-Black trees, B-Trees, min/max binary heaps, hash tables, collision resolution strategies.
- **Algorithms & Analysis:** Asymptotic notation (Big-O, Big-Omega, Big-Theta), divide-and-conquer, greedy algorithms, dynamic programming, sorting algorithms (QuickSort, MergeSort, HeapSort).
- **Graph Theory:** Graph representations (adjacency matrix, list), BFS, DFS, Dijkstra's shortest path, Bellman-Ford, Kruskal's and Prim's minimum spanning trees.

Object-Oriented Programming (CoSc 2021)

Credits: 4 ECTS Equiv: 6 ECTS

Category: Software Development | **Lab:** 3 hrs/wk | **Lecture:** 3 hrs/wk

Advanced object-oriented paradigms, class hierarchies, design patterns, and polymorphic execution in Java and C++.

- **Core Paradigms:** Encapsulation, inheritance, polymorphism, abstract classes, interfaces, generic programming, exception handling, multithreading, memory management.
- **Design Patterns:** Creational (Singleton, Factory), Structural (Adapter, Decorator), and Behavioral (Observer, Strategy) design patterns.

Database Management Systems (CoSc 2042)

Credits: 4 ECTS Equiv: 6 ECTS

Category: Data Management | **Lab:** 3 hrs/wk | **Lecture:** 3 hrs/wk

Relational database theory, normalization algorithms, SQL query optimization, and transaction processing.

- **Theory:** Relational algebra, tuple relational calculus, ER modeling, functional dependencies, normal forms (1NF, 2NF, 3NF, BCNF).
- **Transaction Management:** ACID properties, serializability, two-phase locking (2PL), deadlock detection, write-ahead logging (WAL), crash recovery.
- **Implementation:** Advanced SQL, indexing structures (B+ Trees, Hash indexes), query execution plans, PostgreSQL and MySQL implementations.

Computer Networks and Data Communications (CoSc 3032)

Credits: 4 ECTS Equiv: 6 ECTS

Category: Networking & Telecommunications | **Lab:** 2 hrs/wk | **Lecture:** 3 hrs/wk

In-depth study of network protocol stacks, physical transmission media, routing architectures, and transport reliability.

- **Layer Architectures:** OSI and TCP/IP reference models, framing, error detection (CRC), flow control, ARQ protocols (Go-Back-N, Selective Repeat).
- **Network & Transport:** IPv4/IPv6 addressing, subnetting, CIDR, routing algorithms (Dijkstra, distance vector, BGP, OSPF), TCP congestion control (AIMD, slow start), flow control, UDP sockets.
- **Application Layer:** DNS, HTTP/HTTPS, TLS handshake, SMTP, SSH, network security fundamentals.

3. MATHEMATICS & QUANTITATIVE FOUNDATIONS

Calculus for Computer Science (Math 1011)

Credits: 4 ECTS Equiv: 6 ECTS

Differential and integral calculus: limits, continuity, derivative techniques, Mean Value Theorem, Taylor series, Riemann integration, multivariate partial derivatives, gradients, and optimization.

Discrete Mathematics (Math 1022)

Credits: 3 ECTS Equiv: 5 ECTS

Mathematical logic, propositional and predicate calculus, methods of proof (induction, contradiction), set theory, relations, combinatorics, recurrence relations, Boolean algebra, and formal grammars.

Linear Algebra (Math 2011)

Credits: 3 ECTS Equiv: 5 ECTS

Vector spaces, subspaces, linear independence, basis and dimension, matrices, determinants, Gaussian elimination, eigenvalues, eigenvectors, diagonalization, orthogonalization (Gram-Schmidt), and applications in computer graphics and machine learning.

Probability and Statistics for Computing (Stat 2022)

Credits: 3 ECTS Equiv: 5 ECTS

Probability axioms, conditional probability, Bayes' theorem, discrete and continuous random variables (Binomial, Poisson, Gaussian), expectation, variance, Law of Large Numbers, Central Limit Theorem, hypothesis testing, and regression analysis.

4. SENIOR UNDERGRADUATE CAPSTONE PROJECT

Senior Capstone Research Project (CoSc 4032)

Credits: 4 ECTS Equiv: 8 ECTS

Grade: A (Excellent) | **Duration:** Full Final Academic Year

Title: Design and Implementation of a High-Concurrency Fault-Tolerant Distributed Data Synchronization Protocol

Abstract & Methodology: Architected a lightweight state-synchronization protocol across distributed nodes using Python AsyncIO and Go. Engineered delta-encoded delta serialization to minimize bandwidth consumption across high-latency networks. Implemented heartbeats, leader election routines, and thread-safe lock-free memory rings for high-throughput concurrent I/O.